

Full Academic Syllabus: Artificial Intelligence & Data Science Foundation

1. Mathematical Foundations

Essential for every model, algorithm, and understanding in AI.

• Linear Algebra

- Scalars, Vectors, Matrices, Tensors
- Matrix Operations (Add, Multiply, Transpose, Inverse)
- Determinant, Rank, Trace
- Eigenvalues and Eigenvectors
- Dot Product, Cross Product
- Vector Spaces, Linear Dependence & Independence
- Orthogonality, Projections, Basis

• Calculus

- Limits and Continuity
- Derivatives and Gradients
- Partial Derivatives
- Chain Rule, Product Rule
- Optimization: Minima, Maxima
- Multivariable Functions
- Jacobian and Hessian

• Probability and Statistics

- Probability Theory: Joint, Marginal, Conditional, Bayes Theorem
- Probability Distributions: Uniform, Binomial, Normal, Poisson
- Expected Value, Variance, Covariance, Standard Deviation
- Law of Large Numbers, Central Limit Theorem
- Sampling Techniques
- Hypothesis Testing, p-values, t-test, chi-square test
- Confidence Intervals
- Correlation vs Causation

Assignments:

- Visualize vectors, matrices using Python.
- Compute eigenvalues and vectors by hand.
- Simulate normal distribution.
- Manually derive gradients of simple functions.
- Perform basic hypothesis testing with small samples.

2. Programming & Data Handling in Python

Foundation of implementation, analysis, and automation in DS/AI.

• Core Python Programming

- Variables, Data Types
- Loops, Conditional Statements
- Functions, Scope
- Lists, Tuples, Sets, Dictionaries
- Error Handling
- File I/O
- List Comprehensions
- OOP (Classes, Inheritance, Methods)
- **Python Libraries for Data Science**
 - NumPy: Arrays, Matrix Operations, Broadcasting
 - Pandas: Series, DataFrames, Indexing, Filtering, Merging
 - Data Cleaning: Handling Nulls, Duplicates, Outliers
 - Data Transformation: Encoding, Scaling, Normalizing

Assignments:

- Build your own DataFrame class from scratch.
- Clean a messy dataset (CSV) using pandas.
- Simulate rolling dice & analyze outcomes.
- Write Python code to implement matrix operations.

3. Data Visualization & EDA

Making sense of data visually and statistically.

- **Visualization Libraries**
 - Matplotlib: Lines, Scatter, Bar, Histogram, Pie
 - Seaborn: Pairplot, Heatmap, KDE, Categorical Plots
 - Plotly (optional): Interactive visualizations
- **Exploratory Data Analysis**
 - Univariate Analysis
 - Bivariate and Multivariate Relationships
 - Correlation Matrix and Feature Analysis
 - Grouping and Aggregation

Assignments:

- Create heatmap for correlation on real-world dataset.
- Plot pairplot of Iris Dataset.
- Perform EDA on Titanic dataset or any dataset from Kaggle.

4. Core Machine Learning (Classical ML)

The true heart of AI foundations before deep learning.

- **ML Foundations**
 - What is Machine Learning?
 - Supervised vs Unsupervised vs Reinforcement Learning
 - ML Lifecycle (Data > Features > Model > Evaluate > Deploy)
 - Overfitting, Underfitting, Bias-Variance Tradeoff
- **Supervised Learning**

- Linear Regression
- Polynomial Regression
- Logistic Regression
- k-Nearest Neighbors (k-NN)
- Naive Bayes
- Decision Trees
- **Unsupervised Learning**
 - Clustering: K-Means, Hierarchical
 - Dimensionality Reduction: PCA
- **Model Evaluation**
 - Regression: MSE, MAE, RMSE, R2 Score
 - Classification: Accuracy, Precision, Recall, F1 Score
 - Confusion Matrix, ROC Curve, AUC
 - Cross Validation

Assignments:

- Implement Linear & Logistic Regression from scratch.
- Build k-NN classifier without libraries.
- Analyze bias-variance using plots.
- Cluster a simple dataset using K-Means (2D visualization).

5. Feature Engineering & ML Enhancements

Improving models with better data understanding.

- **Feature Engineering**
 - Feature Extraction vs Feature Selection
 - Encoding Categorical Data
 - Handling Imbalanced Data
 - Creating New Features from Date/Time/Text
 - Feature Scaling: Normalization, Standardization
- **Ensemble Learning**
 - Bagging: Random Forest
 - Boosting: Gradient Boosting, AdaBoost, XGBoost (intro)
 - Stacking
- **Hyperparameter Tuning**
 - Grid Search
 - Randomized Search
 - Manual tuning with validation curve

Assignments:

- Create new features from a time series dataset.
- Compare Random Forest vs Decision Tree on same dataset.
- Run a GridSearchCV on a classification model.

6. Time Series Analysis

Understanding data over time and forecasting.

- **Time Series Fundamentals**

- Stationarity
- Lag, Rolling Statistics, Moving Average
- Seasonality, Trend
- Decomposition

- **Time Series Models**

- Autocorrelation (ACF), Partial Autocorrelation (PACF)
- AR, MA, ARIMA Models
- Seasonal ARIMA (SARIMA)

Assignments:

- Plot moving averages of temperature data.
- Forecast future sales using ARIMA.
- Decompose a time series using statsmodels.

7. Natural Language Processing (NLP)

How machines read, process, and understand text.

- **Text Preprocessing**

- Lowercasing, Tokenization
- Stopword Removal
- Lemmatization/Stemming

- **Text Representation**

- Bag of Words
- TF-IDF
- Word Embeddings (Word2Vec, GloVe - conceptual)

- **Text Classification**

- Naive Bayes Classifier
- Logistic Regression on text data
- Sentiment Analysis (intro)

Assignments:

- Build a spam classifier.
- Compare BoW and TF-IDF representations.
- Visualize word frequencies.

8. Deep Learning Foundations

The fundamentals of neural computation.

- **Neural Network Basics**

- Perceptron, Multi-Layer Perceptron (MLP)
- Activation Functions: Sigmoid, Tanh, ReLU
- Forward and Backward Propagation
- Loss Functions: MSE, Cross Entropy
- Optimizers: Gradient Descent, SGD, Adam

- **Regularization**

- Dropout
- L1/L2 Regularization

- Batch Normalization
- **CNN Basics**
 - Convolutional Filters, Feature Maps
 - Pooling Layers
 - Padding, Stride
- **RNN Basics (Intro Only)**
 - Sequential Data, Memory, Timesteps
 - Vanishing Gradient Problem

Libraries

- PyTorch / TensorFlow / Keras Basics

Assignments:

- Build MLP on MNIST dataset.
- Visualize filters in a CNN.
- Implement a basic RNN to predict character sequences.

Optional Advanced Concepts (Cover if time allows)

- Information Theory: Entropy, KL Divergence
- Reinforcement Learning Basics (Markov Decision Process, Q-Learning)
- Graph Theory & NetworkX
- Probabilistic Graphical Models
- Generative Models (GANs, VAEs)

Recommended Tools & Resources

- **Jupyter / Google Colab** (for experiments)
- **Scikit-learn, Statsmodels, TensorFlow/Keras/PyTorch**
- **Matplotlib, Seaborn, Plotly**
- **Desmos, WolframAlpha** (for math)
- **Kaggle, UCI ML Repo** (for datasets)
- **Books:**
 - Mathematics for ML (Marc Deisenroth)
 - Hands-On ML with Scikit-Learn, Keras & TensorFlow (A. Geron)